

Research Report

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Introduction

Preregistrations were popularized in the 2010s in psychology as a way to increase reproducibility (Lindsay & Stephen, 2018a). Preregistrations are documents that are published at the start of the research process, in which decisions concerning data collection and analysis are recorded. This is done with the aim to limit the amount of choices a researcher can make during their project, thereby limiting how much can be steered towards desired results.

This method of increased transparency about research choices has become more popular over the years with the number of preregistrations increasing yearly (Ferguson et al., 2023; Lindsay & Stephen, 2018b). However, whether the desired effect of limiting flexibility and improving reproducibility has been achieved is debated. It is often argued that preregistrations should lead to more credible and reproducible results (Lakens, n.d.-a; Nosek et al., 2018; Wagenmakers et al., 2012). However, research has also shown no difference between preregistered and non-preregistered papers in the number of significant findings (Akker et al., 2024a), nor a reduction in p-hacking (Brodeur et al., 2024).

There are many reasons for the uncertainty of the effect. Preregistrations are often incomplete or unspecific about their decisions (Claesen et al., 2021; Heirene et al., 2024). Furthermore, researchers also deviate from their predetermined choices quite often (Claesen et al., 2021; Van Den Akker et al., 2024). Deviations have been found in as high as 93% of preregistered papers, with as much as 63% going unreported (Claesen et al., 2021; Willroth & Atherton, 2024). Little is known about the effects of these divergences.

The aim of this project is to identify problematic deviations and investigate their effects. I first go into more depth about the goals and problems of preregistrations and the potential impact of deviations. Based on how commonly they occur, how problematic they are in theory, and how justifiable they are, I identify multiple deviation domains to investigate. Then, through a simulation study I examine the effect of said deviations on the type I and type II error rates in research **and evaluate what this means for the existing literature.**

The Problem and the Solution

The inability to reliably reproduce findings in the field of psychology is a known problem. In 2015, (Open Science Collaboration, 2015) showed that only 36%

of significant results could be reproduced. One reason for this replication crisis is ‘researcher degrees of freedom’ (Simmons et al., 2011). This term describes the flexibility that researchers have in decisions made during their work. This includes decisions such as how many participants to collect, which statistical model to use and which covariates to add. When these decisions are made during data collection or analysis, researchers can make opportunistic choices to steer their results towards the desired outcome. This action is called ‘p-hacking’ and can dramatically increase false-positive rates of research (Stefan & Schönbrodt, 2023).

In order to limit these types of practices, preregistrations, amongst other solutions, were proposed. Decisions concerning the research process are documented before data collection in order to limit flexibility during the research process. In theory this works well, if researchers are specific in their preregistrations and follow them closely, this should lead to better reproducible findings. In practice, this happens differently. Preregistrations are often imprecise and incomplete, meaning that their limitation of flexibility is diminished. Furthermore, deviations from preregistrations also happen commonly (Claesen et al., 2021; Van Den Akker et al., 2024). Meaning that during the actual research, predetermined plans are not being followed. Changes between preregistrations and final publications occur most commonly in data collection procedures, statistical models and exclusion criteria (Van Den Akker et al., 2024). This includes discrepancies in the total sample size, how outliers are selected and which covariates are used. This could serve to further diminish the effectiveness of preregistrations, as it reintroduces researcher degrees of freedom and leaves room to revert back to p-hacking strategies.

The good, the bad, and the unknown / Are all deviations bad? / The problem of the solution

Deviations from preregistrations occur in all different aspects of the research process, but what is not known is the effect of these divergences. Do these changes from original research plans negatively impact the quality of the research? Lakens (2024) theorized what the possible consequences could be. He argues that if deviations occur due to assumptions being violated or data no longer being suitable for the planned analysis, then deviating from the original research plan can have positive effects on the validity of a test. Similarly, adding additional analyses can increase the robustness. On the other hand, changes in testing also mean that a test is generally less severely tested. Meaning that a theory has not been given the proper opportunity to be falsified which can result in higher type II errors. Consequently, little can be said about whether deviations are bad, especially when the reason for the deviation is unknown.

When evaluating whether inconsistencies between preregistrations and final publications are problematic, researchers themselves are also uncertain. When asked by Willroth & Atherton (2024), researchers deemed certain deviations to be more problematic than others. No deviation is considered to be very acceptable but changes to analyses and hypotheses are deemed least acceptable,

whereas changing the research platform or using different software might be justifiable. These judgements were based not only on the type of deviation but also on whether the deviation was reported transparently and what the estimated impact on the results of the study was. However, as mentioned, assessing the impact of these deviations remains difficult.

This project

With this simulation study, I intend to close this gap in the literature and explore the effects of typical preregistration deviations on research outcomes and the quality of published results. Specifically, I aim to answer the question: ‘How do deviations from preregistration affect Type I and Type II error rates in research?’. The effect of divergences is studied in three domains: sample size, outlier identification, and the statistical model. Different deviations are simulated in each of these domains and compared to a baseline condition with no deviations. This project investigates this question in the context of behavioral psychology and uses linear regression as the main model of interest. The effects are examined across two scenarios: a no-effect scenario, and an effect scenario. The aim of this study is descriptive and therefore no specific hypotheses are formalized.

As a secondary goal, this project also addresses the issue of reasons for deviations being unknown. Each deviation condition is initially simulated as random and the outcomes will be used to assess type I and type II error. Next, deviations will be simulated and compared to the baseline condition, results will be chosen opportunistically from either condition after which error rates will then be compared.

With these simulations I hope to provide clarity on the effect of deviations from preregistration on type I and type II errors, as well as shed more light on the effect of picking deviations opportunistically.

Methods

Aims

The aim of this simulation study is to investigate the impact of making changes to analysis plan on the type I and type II error rates.

Conditions

I identified important deviations to investigate based on (1) how often they occur, (2) their potential impact and (3) their justifiability. These deviations occur in the following domains: sample size, outlier identification, and statistical model. Within these domains, multiple conditions will be simulated and compared to a baseline condition without any deviations.

Sample size

Deviations in the sampling size are some of the most common deviations, with consistency between preregistrations and published papers only being 28% for the exact sampling size (Akker et al., 2024b). Changing the sample size can decrease power in research as well as inflate the false-positive rates (Lakens, n.d.-b; Simmons et al., 2011). Small deviations in sample size are common and often due to oversights during the collection process. Researchers also admit to stopping collection earlier or continuing collection longer after finding disappointing results (John et al., 2012), however, whether this is also used as a reason to deviate from preregistrations is unknown. Based on this, the conditions in Table 1 are simulated within the sample size domain.

Outlier identification

Literature shows that over half of published papers do not adhere to their own specified outlier criteria (Claesen et al., 2021; Van Den Akker et al., 2024). Reasons for these deviations often include vague criteria in the preregistration or failing to specify outlier criteria at all in preregistrations. When criteria are specified, they often leave a lot of room for interpretation and ad hoc decisions (Heirene et al., 2024). This could potentially lead to researchers choosing how to exclude outliers based on which method provides better results. However, Bakker & Wicherts (2014) showed that there is no significant difference in the amount of significant findings between papers with outlier exclusion and those without. This suggests that adding outlier exclusion criteria when they were not specified in the preregistration might not be a problem. Researchers also deem outlier deviations to be relatively acceptable (Willroth & Atherton, 2024).

However, Lakens (n.d.-b) argues that adding additional exclusion criteria can undermine the severity of a test. This idea is corroborated by Simmons et al. (2011), who showed that the Type I error increases linearly with the number of outlier methods used. Furthermore, 38% of researchers admit to excluding outliers only after examining their impact on the data (John et al., 2012). If this is also the reason why many people deviate from their preregistered outlier criteria, then these deviations could be impactful. Consequently, whilst deviations in outlier identification are common and relatively accepted, they might have quite an effect.

The most common method for excluding outliers in the social sciences is the Z-score. A minimum of three standard deviations is often maintained as the rule of thumb. Within linear regression, it is also common to base outliers on influence. One of the more commonly used methods is Cook’s distance. Based on this, the baseline condition will be to not exclude any datapoints, with deviation conditions based on z-score and Cook’s distance Table 1.

Statistical model

Amongst all of the aforementioned domains, the least is known about why people deviate from their preregistered model. Many areas to exist within the statistical model in which researchers can choose to change their predetermined plans. Including, but not limited to, changes in the dependent variable, in-

dependent variable, statistical inference criteria and the actual model that is assessed. Different research have investigated deviations in this domain differently. Claesen et al. (2021) shows a divergence rate of 70% within the ‘analysis’ domain, of which most divergences go unreported. Whereas Van Den Akker et al. (2024) shows inconsistencies in 40% of the statistical models.

As Lakens (n.d.-b) argues, at times it might be necessary to alter the statistical model used. For example, when a variable has been measured at a different level than expected or when data from two different indicators cannot be combined. However, changing the statistical model can also be done opportunistically. Adding an extra dependent variable can increase the type I error rate to 9.5%, and the addition of a covariate can increase it to 11.7% (Simmons et al., 2011). Therefore, the reasoning behind the change becomes very important. Deviations due to failing to properly specify how variables would be operationalized in the preregistration are therefore also seen as a larger shortcoming than model deviations due to unforeseen consequences (Willroth & Atherton, 2024). Within the statistical model, three deviation conditions are simulated; outcome switching, adding a continuous covariate, and adding a dichotomous covariate (Table 1.)

Table 1: Deviation Domains and Corresponding Deviation Conditions

Domain	Simulation Condition
Baseline	No deviations
Sample Size	+5 participants
	+30 participants
	-5 participants
	-30 participants
Outlier Identification	Z-score of $> 3 sd$
	Cook’s distance of > 1
Statistical Model	Outcome Switching
	Adding a continuous covariate
	Adding a dichotomous covariate

Data-generating mechanism

Data is generated under two scenarios; the hypothesis is true and the hypothesis is not true. The hypothesis is as follows: ‘X has an effect on Y’. Both scenarios generate data by producing parametric draws from a linear regression model.

$$Y = \beta_0 + \beta_1 X + \epsilon$$

. In which y is the dependent variable, β_0 the intercept, β_1 the main effect, and ϵ the random error. Therefore the one difference in data generation between the two scenarios is in the β_1 parameter. In the $H = true$ scenario, β_1 is 0.3, and in the $H = false$ scenario, β_1 will be 0. In order to asses differences between outlier identification methods, the data need to include outliers. In

order to simulate outliers, 5% of the data is simulated with an increased or decreased value for ϵ . The parameter values for the data generating mechanism are specified in Table 2.

Table 2: Data generating mechanism

Parameter	Value
Intercept	$\beta_0 = 0$
Regression coefficient	$\beta_1 = 0$ or $\beta_1 = 0.30$
Independent variable	$X \sim \mathcal{N}(\mu, \sigma^2)$
Random error	95% = $\epsilon \sim \mathcal{N}(0, 1)$
	2.5% = $\epsilon \sim \mathcal{N}(-2, 1)$
	2.5% = $\epsilon \sim \mathcal{N}(2, 1)$
Continuous demographic variable	$Z \sim \mathcal{N}(\mu, \sigma^2)$
Dichotomous demographic variable	$D \sim \text{Bernoulli}(0.5)$
Sample size	200

Estimands

Each condition will be assessed on how they affect the type I and type II error. The type I error refers to a false-positive, or rejecting the null-hypothesis when it is true. In this study that would mean detecting an effect in the no-effect scenario. Type I error is generally acceptable at a rate of 5% or below, based on an alpha level of 0.05. In this case, a rate of higher than 5% is considered an inflated type I error rate.

Type II error refers to a false negative, or failing to reject the null-hypothesis when it is false. In this study that would mean failing to detect an effect in the effect scenario. Type II error is commonly assessed in the form of power. Power is $1 - \beta$, where β is the type II error. The nominal type II error rate of 0.20 results in a generally accepted power level of 80%. Consequently, in this study, power rates of below 80% are considered an inflated type II error rate.

Methods

As previously mentioned, there are two scenarios under which data is generated. A no-effect scenario, in which the hypothesis is false and an effect scenario, in which the hypothesis is true. Within these scenarios each simulation is examined under different conditions. The conditions include a baseline condition, with no deviations, and one condition for each possible deviation (#tbl-cond)

Table 3: Parameter Values for Baseline and Deviation Conditions

Parameter	Baseline condition	Deviations conditions
Sample size	200	+5 +30 -5 -30
Outliers	None	Based on Z-scores Based on Cook's distance
Covariates	None	Continuous covariate Dichotomous covariate
Outcome variable	Y	Ya

If time allows, After examining the effects per condition, the effect of choosing whether or not to deviate is examined. This is done by simulating each deviation condition as reported in #tbl-cond, but instead of automatically examining the value of the deviation condition, a choice is made. In each iteration the p-value of the deviation condition is compared to the p-value of the baseline condition, the most desired p-value is added reported as the outcome. An overview can be found in Figure (**fig:process-phases?**).

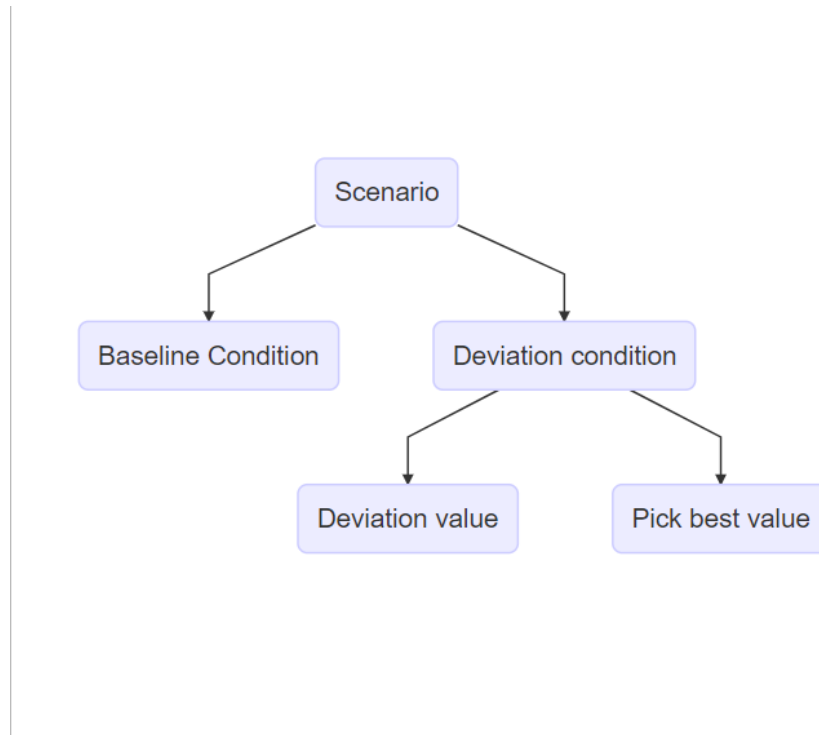


Figure 1: Process Flowchart

Performance measures

Performance is assessed through the Type I and Type II error. Furthermore, the regression coefficient and confidence interval are collected as well.

Results

- include what I expect results to look like

Discussion

- limitation of specific field
- what does this research say about the field and what has been published till now
- how do we keep the field reliable? which deviations should we try to avoid and which ones do not seem to matter much?
- hypotheses so many deviations, could be concern but outside of this scope

Conclusion

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